# ECOLINK – HYPERLOCAL MARKETPLACE FOR SUSTAINABLE PLANTS, ECO-FRIENDLY PRODUCTS & COMMUNITY

**23CS1ME – MINI CAPSTONE PROJECT**

*Submitted by*

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# BONAFIDE CERTIFICATE

This is to certify that this project report, "EcoLink – Hyperlocal Marketplace for Sustainable Plants, Eco-Friendly Products & Community", is the Bonafide work of **JAYA SHIVAANI K P (2312122)** who carried out the project work under my supervision.

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# ABSTRACT

The rapid growth of centralized e-commerce platforms has drastically expanded consumer choice, yet it has inadvertently created severe environmental and socioeconomic challenges. Long-distance supply chains for living nursery plants and perishable farm produce result in excessive packaging waste, severe transit mortality of delicate saplings, and heightened carbon emissions from motorized logistics. Simultaneously, micro-scale urban gardeners, rural artisans, Self-Help Groups (SHGs), and marginalized women entrepreneurs struggle to compete against mass-manufactured goods due to exorbitant distributor markups, complicated onboarding procedures, and high listing commissions on mainstream digital marketplaces. In parallel, urban communities lack a dedicated, localized mechanism to redistribute surplus garden saplings, exchange rare botanical cuttings, or organize green community workshops.

To address these systemic supply chain and community fragmentation issues, EcoLink introduces a full-stack, software-first Hyperlocal Marketplace and Sustainable Community Ecosystem. Developed using the MERN stack (MongoDB, Express.js, React.js, and Node.js), EcoLink integrates high-precision proximity discovery using the Browser Geolocation API, interactive geospatial rendering via Leaflet.js / OpenStreetMap, and accurate spatial filtering powered by the Haversine distance algorithm (5 km to 50 km adjustable radius). The platform incorporates a unified Multi-Role Account architecture, enabling registered Buyers to seamlessly apply for Seller privileges through a rigorous admin verification workflow. Verified Self-Help Groups (SHGs) and Women Entrepreneurs are supported through dynamic, configurable commission structures and distinct trust badges. Beyond commerce, EcoLink features an innovative "Plant Loop" comprising a zero-cost Plant Adoption Board, a circular Plant Exchange Hub, and a corporate Bulk Business Requirement Bidding Portal. Secure financial transactions are conducted via an integrated Razorpay payment gateway with cryptographic HMAC-SHA256 signature verification, while dynamic localized interfaces (English and Tamil) ensure complete accessibility. EcoLink delivers an end-to-end, resilient digital ecosystem that fosters neighborhood self-reliance, circular gardening, and green economic empowerment.

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# CHAPTER 1

# INTRODUCTION

In an era defined by rapid urbanization and mounting ecological concerns, the demand for green living, urban gardening, and sustainable consumer goods has witnessed unprecedented expansion. Modern consumers actively seek indoor plants, medicinal herbs, organic fertilizers, and zero-waste home alternatives. However, the existing digital retail landscape remains heavily centralized. Mainstream e-commerce giants operate on continental warehousing networks that necessitate extensive freight transport, multiple transit hubs, and heavy synthetic packaging. For delicate living botanical products and fresh agricultural goods, long-distance transit leads to high plant mortality rates, severe transit shock, excessive plastic cushioning waste, and avoidable greenhouse gas emissions.

Simultaneously, grassroots producers—including home-scale nursery owners, rural artisans, organic smallholders, Self-Help Groups (SHGs), and independent women entrepreneurs—face prohibitive barriers to digital commerce. Centralized platforms impose steep onboarding fees, rigid volumetric logistics agreements, and punitive commission structures ranging between 15% and 35%. Consequently, localized green producers are economically marginalized and unable to showcase their specialized organic goods to immediate neighborhood consumers. In parallel, urban gardening enthusiasts possess surplus saplings, seed harvests, and propagated cuttings but lack an organized local forum to donate or exchange them, leading to horticultural wastage.

To resolve these systemic inefficiencies, EcoLink establishes an integrated, software-first Hyperlocal Marketplace and Circular Community Platform. Built on the modern MERN stack (MongoDB, Express.js, React.js, and Node.js), EcoLink anchors digital trade within immediate geographical vicinities. By harnessing the browser's native Geolocation API and open-source Leaflet.js mapping, the system calculates precise user-to-seller distances using the Haversine formula, enabling consumers to discover and procure sustainable goods within customizable radii (5 km to 50 km).

## 1.1 PROBLEM STATEMENT

The primary purpose of the EcoLink Hyperlocal Marketplace and Community Ecosystem is to bridge the divide between local sustainable producers and eco-conscious neighborhood consumers while eliminating the carbon-intensive overhead of traditional e-commerce. Conventional online platforms treat living plants and artisan eco-goods as standard dry freight, resulting in high transit damage, excessive single-use plastic waste, and significant transit carbon footprints. Furthermore, micro-producers, rural artisans, SHGs, and women entrepreneurs are frequently excluded from mainstream digital visibility due to high commissions and complex onboarding procedures. Traditional systems also fail to facilitate community-driven circularity, such as plant adoption, seed swapping, or transparent bulk business bidding between local institutions and suppliers. EcoLink addresses these critical challenges by providing a location-aware, multi-role web platform that supports dynamic distance filtering via the Haversine formula, administrative seller verification, role-specific commission structures, circular plant re-homing, and secure digital payments.

## 1.2 SCOPE OF THE PROJECT

The scope of this project encompasses the complete engineering, architectural design, and deployment of a full-stack, responsive web application utilizing the MERN stack (MongoDB, Express.js, React.js with Vite, and Node.js). Key technical features include multi-role user management, administrative verification queues, Leaflet.js / Haversine distance filtering (5 km–50 km), dynamic multi-category botanical

cataloging, circular Plant Loop (Adoption & Exchange), institutional bulk tender bidding, and Razorpay payment gateway integration with HMAC-SHA256 signature verification.

# CHAPTER 2

# LITERATURE REVIEW

The continuous growth of digital marketplaces has transformed global retail, prompting extensive academic inquiry into short-chain food systems, decentralized e-commerce, and hyperlocal logistics. The following survey examines key advancements across academic publications and journals to establish the foundational research gap addressed by EcoLink.

**2.1 Hyperlocal E-Commerce and Short-Chain Logistics:** Sharma and Kumar (2022) examined the operational dynamics of urban micro-warehousing and hyperlocal delivery networks for perishable goods. Their study demonstrated that restricting delivery corridors to a 15 km radius reduced logistics carbon emissions by 42% and minimized packaging material waste.

**2.2 Community-Driven Botanical Exchange & Circular Economy:** Patel et al. (2023) investigated peer-to-peer botanical sharing networks across metropolitan communities, demonstrating that informal plant swapping significantly increases urban biodiversity and reduces domestic organic waste.

**2.3 Trust Verification & Marginalized Producer Empowerment:** Sundaram et al. (2024) evaluated the barriers faced by women-led cottage industries when entering mainstream e-commerce platforms. The authors identified rigid onboarding criteria and uniform 20%+ commission fees as the primary causes of high seller churn.

**2.4 Geospatial Proximity Algorithms in Web Applications:** Alvarez and Jenkins (2025) analyzed spatial indexing mechanisms, showing that for lightweight web applications operating on shared hosting environments, computing trigonometric spherical distances via the Haversine formula provides high precision with low overhead.

**2.5 Research Gap:** Commercial e-commerce platforms ignore the delicate shipping constraints of living plants and impose prohibitive commissions on micro-producers. Conversely, community plant forums lack secure checkout, distance calculations, and seller vetting. EcoLink directly bridges this gap with an integrated MERN ecosystem.

# CHAPTER 3

# DESIGN THINKING

## 3.1 EMPATHY MAP

The Empathy Map is utilized to analyze user behaviors, emotions, and expectations across urban plant buyers, local green sellers / SHG members, and platform administrators.

**EMPATHY MAP SUMMARY**

- **Says:** 'Can I buy fresh saplings locally?' | 'I want to donate extra cuttings.'
- **Thinks:** 'Will long-distance transit kill delicate indoor plants?' | 'High commissions hurt small sellers.'
- **Does:** Searches local nurseries, swaps plant babies, seeks verified SHG goods.
- **Feels:** Frustrated by packaging waste and plant transit shock; empowered by supporting local women.

## 3.2 STAKEHOLDER ANALYSIS & PAIN POINTS

Key stakeholders include Buyers (seeking fresh local plants and eco-goods), Sellers (local nurseries, SHGs, and women artisans), Institutional Buyers (schools and offices tendering bulk requirements), and Admins (auditing credentials and commission percentages). Primary pain points resolved include plant transit mortality, prohibitive 20%+ commissions, and lack of neighborhood plant re-homing.

## 3.3 DESIGN GOALS & FEASIBILITY

Goals include delivering 5 km–50 km Haversine proximity filtering, single-account multi-role expansion upon admin vetting, SHG/Women badges with preferential commission rates, zero-cost plant adoption, and cryptographic Razorpay checkout.

# CHAPTER 4

# PROPOSED METHODOLOGY

## 4.1 SYSTEM ARCHITECTURE & DESIGN

EcoLink employs a decoupled MERN architecture with an Express.js API gateway, JWT stateful authentication, Leaflet.js interactive maps, and MongoDB Mongoose persistence.

**EcoLink Multi-Tier Architecture & Data Flow:**
1. **Client Tier:** React.js 18 (Vite), Tailwind CSS, Leaflet.js Maps, Axios.
2. **API Gateway:** JWT Authentication Middleware, Role RBAC, Multer file upload.
3. **Application Logic:** Node.js / Express controllers for Products, Orders, Bids, Plant Loop.
4. **Persistence & External:** MongoDB (Mongoose ODM), Razorpay API (HMAC-SHA256).

## 4.2 ALGORITHMIC STEPS

• **Haversine Distance Formula:** Spherical distance calculation between buyer coordinates (lat1, lon1) and seller coordinates (lat2, lon2) within 5 km to 50 km.
• **Seller Verification Pipeline:** Buyer submits business profile and proof document; admin audits queue; approval activates dual Buyer + Seller account.
• **Payment & Commission Engine:** Dynamic commission computation and Razorpay HMAC-SHA256 cryptographic signature verification.

## 4.3 SYSTEM REQUIREMENTS

| Category | Specification | Purpose |
|---|---|---|
| Processor | Intel i3 / Ryzen 3+ | API & spatial calculation execution. |
| Memory / Storage | 4 GB RAM / 500 MB Space | Node.js runtime & local MongoDB. |
| Software Stack | MERN Stack (MongoDB, Express, React, Node) | Full-stack web application development. |
| Payment Gateway | Razorpay API (Test Mode) | Encrypted digital payments & HMAC validation. |

# CHAPTER 5

# IMPLEMENTATION AND RESULT

## 5.1 IMPLEMENTED MODULES

• **Admin Command Center:** Real-time GMV metrics, seller verification queue, dynamic commission configuration, and role governance.
• **Buyer & Marketplace Module:** Leaflet map proximity discovery (5 km–50 km), botanical classification (decorative, medicinal, yield), itemized cart, and Razorpay Test Mode checkout.
• **Circular Plant Loop:** Plant Adoption Board for zero-cost sapling donations and Plant Exchange Hub for cuttings bartering.
• **Business Bidding Portal:** Bulk institutional tenders with competitive seller quotation submissions.
• **Direct Chat & Notifications:** Full chat page with inbox conversations and active message window fetched via 3-second polling; notification dropdown in the top navigation listing plant adoption, exchange, and order updates with instant contact actions.
• **Multilingual Localization:** Dual-language toggle translating the complete marketplace into Tamil.

## 5.2 RESULTS AND EVALUATION

The system successfully delivers sub-second Haversine proximity calculations within 5 km to 50 km, seamless single-account role transitions from Buyer to Seller upon admin audit, 100% cryptographic integrity in Razorpay payments, and reliable state tracking across all community and marketplace workflows.

# CHAPTER 6

# CONCLUSION AND FUTURE SCOPE

## 6.1 CONCLUSION

The EcoLink – Hyperlocal Marketplace for Sustainable Plants, Eco-Friendly Products & Community has been designed and implemented using the MERN stack to eliminate long-haul transit damage, reduce single-use plastic waste, empower marginalized producers (SHGs and women entrepreneurs), and foster neighborhood botanical circularity.

## 6.2 FUTURE SCOPE

- Progressive Web App (PWA) / mobile expansion with push notifications.
- AI computer vision models for plant leaf health diagnosis.
- Automated hyperlocal courier dispatch integration (ONDC / Dunzo APIs).
- Blockchain carbon credit verification and regional dialect voice navigation.

# CHAPTER 7

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